

S.P.M College, Udantpuri

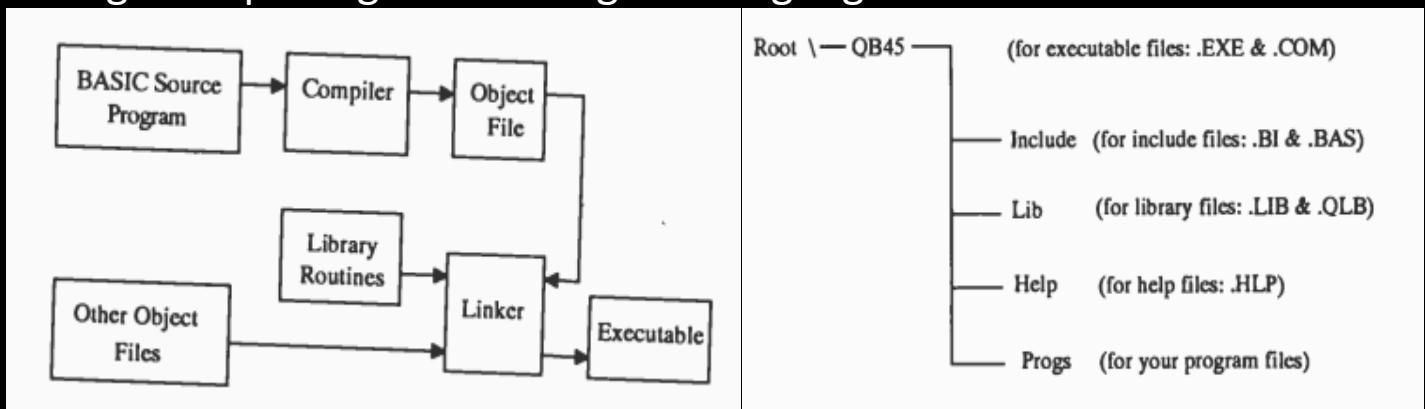
Bachelor Of Computer Application (BCA)

Part -1 (Paper-1)

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QBASIC

- The QBASIC stands for **Quick Beginner's All-purpose Symbolic Instruction Code** and was first invented by Professors J. G. Kemeny and T.E. Kurtz of Dartmouth college, U.S.A. in 1964.
- Version of Basic Q-Basic, GWBASIC, Turbo BASIC, Visual Basic etc.
- As the name suggests right from the beginning BASIC was designed to be a beginner's language.
- It is very user-friendly English like language.
- File extension of qbasic is .BAS or .BI
- This language is not only easy to understand but you can also develop customized software packages for business application, music and games packages etc. using this language.



❖ FEATURES OF QBASIC

1. It is a user friendly language.
2. It is widely known and accepted programming language.
3. It is one of the most flexible languages, as modification can easily be done in already existing program.
4. Language is easy since the variables can be named easily and uses simple English phrases with mathematical expressions.

❖ QBASIC DATA

- Data is a collection of facts and figures that is entered into the computer through the keyboard.

Data is of two types:- 1. Constant 2. Variable

1. Constant - Data whose value does not change or remains fixed.

There are two types of constants:-

i. **NUMERIC CONSTANT:**

- Numbers - negative or positive used for mathematical calculations
e.g. 16, -10, 20, 0, 24.56, -32.20

ii. **ALPHANUMERIC CONSTANT / STRING Constant:-**

- A set of characters is called a String. Numbers or alphabets written within double quotes (inverted commas " ").
- Example, "Computer", "Operating System", "SUM = 44", "162", are some valid string constants.

2. VARIABLE:

- Data whose value is Change and may change due to some calculation during the program execution.
- It is a location in the computer's memory, which stores the values. Depending on what value is held.

Variables are of two types:-

i. **NUMERIC VARIABLE:**

- The variable that holds a Numeric Constant for arithmetic calculations (+, -, *, /) is called a Numeric Variable.
- E.g. A = 50, here A is the Numeric Variable

ii. **ALPHANUMERIC VARIABLE:**

- The variable that holds an Alphanumeric Constant, which cannot be used for arithmetic calculations, is called Alphanumeric Variable or String Variable.
- An Alphanumeric variable must end with a \$ (Doller) sign and the Alphanumeric constant must be enclosed in inverted commas.
- E.g. Name\$ = "Akanksha", here Name\$ is an Alphanumeric Variable

- Example, A2\$ (String Variable) and "tom" (String Constant)
- Example - 2degree (wrong) because starting first alpha than digit
- Example a2 (Right)

Correct the following expressions:

B\$ =WORLD CUP 2007

"DELHI PUBLIC SCHOOL" = D\$

\$\$A = "ARYA PUBLISHING COMPANY"

P\$ = (A + B)2

4B\$ = "NEW DELHI"

A= 4P + 5Q

AVG = (2A + 3B + 4C) / 3

A * B = AB + CD

❖ What is QBASIC Operator?

- Operators in Qbasic are use to perform different mathematical, logical and comparison operations.

There are three types of operators in QBasic:

1.Arithmetic Operators:- (+,-,*,/,^ -Power)

2.Relational Operators :- (=,<>,>,<,>=,<=, BEDMAS Rule, Where E-Exponentiation i.e.Power)

3.Logical Operators :- AND,OR,NOT

- ✓ **Statements :-** The commands or instructions in QBASIC are called statements
- ✓ **Program :-** The collection of related statements to get a desired output is known as program.

Types of statements and their commands of Qbasic:

Type of Statement	Command
Clearing the Output window	CLS (clear)
Assignment Statement	LET
Input Statement	INPUT

Output Statement

PRINT

Remark Statement

REM

Declare Variables/arrays

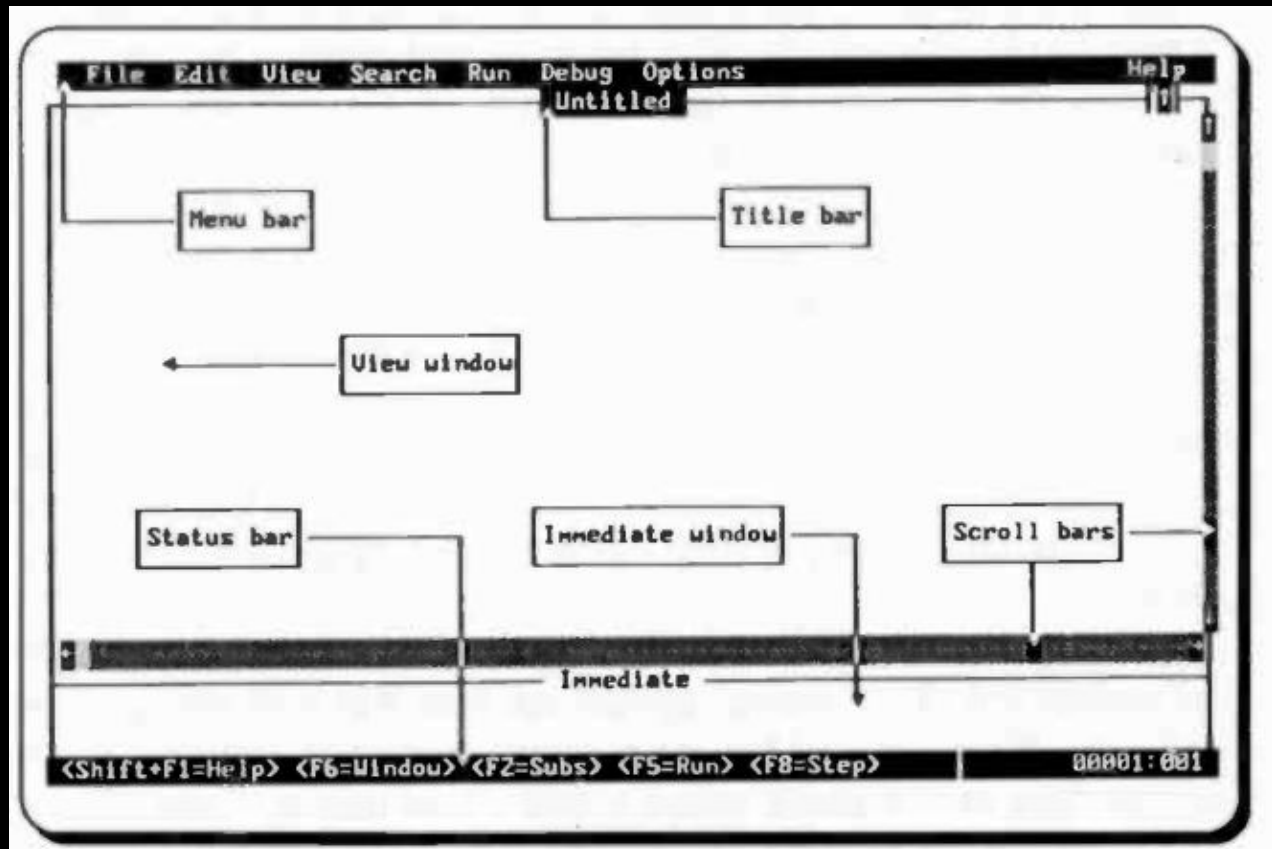
DIM (dimension)

Conditional Statement

IF THEN ELSE

Looping Statement

FOR – NEXT , WHILE-WEND

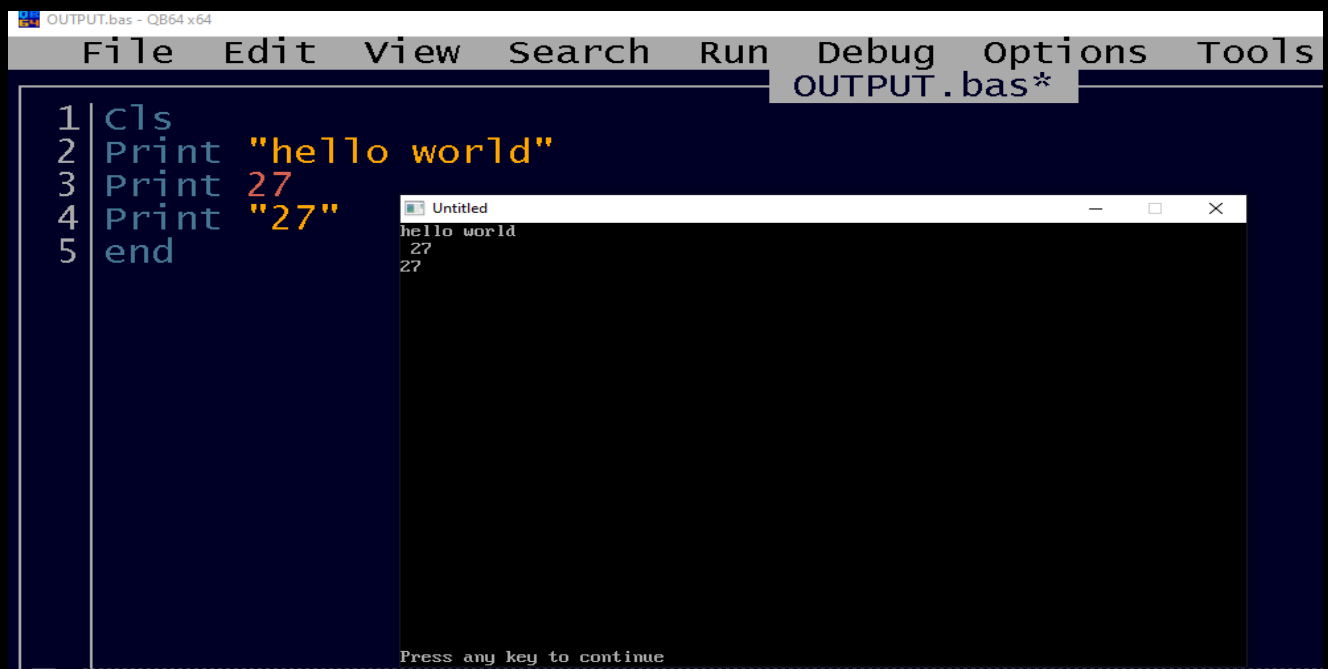


1. Cls Command

- Used to clear the previous output displayed on the screen.
- Syntax :- CLS

2. Print Command

- Used to output the program.
- Print command always used in double quotes(" ").
- Syntax:- PRINT " "
- Example :- Print "Hello World" => Output – Hello World
 Print "20" OR Print 20 => Output – 20



```
1 Cls
2 Print "hello world"
3 Print 27
4 Print "27"
5 end
```

Press any key to continue

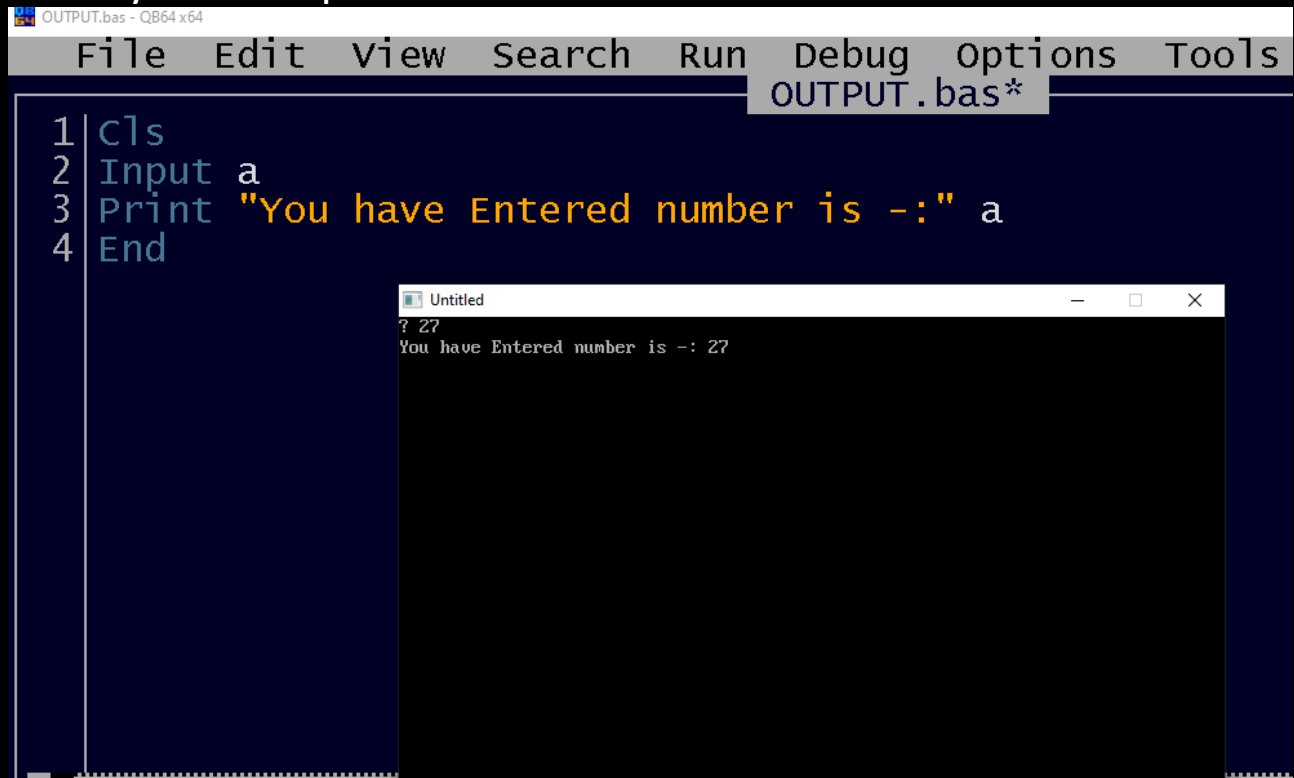
OUTPUT.bas*

Untitled

hello world
27
27

3. INPUT Command

- This command to use input data by user.
- Syntax :- Input a



```
1 Cls
2 Input a
3 Print "You have Entered number is -:" a
4 End
```

OUTPUT.bas*

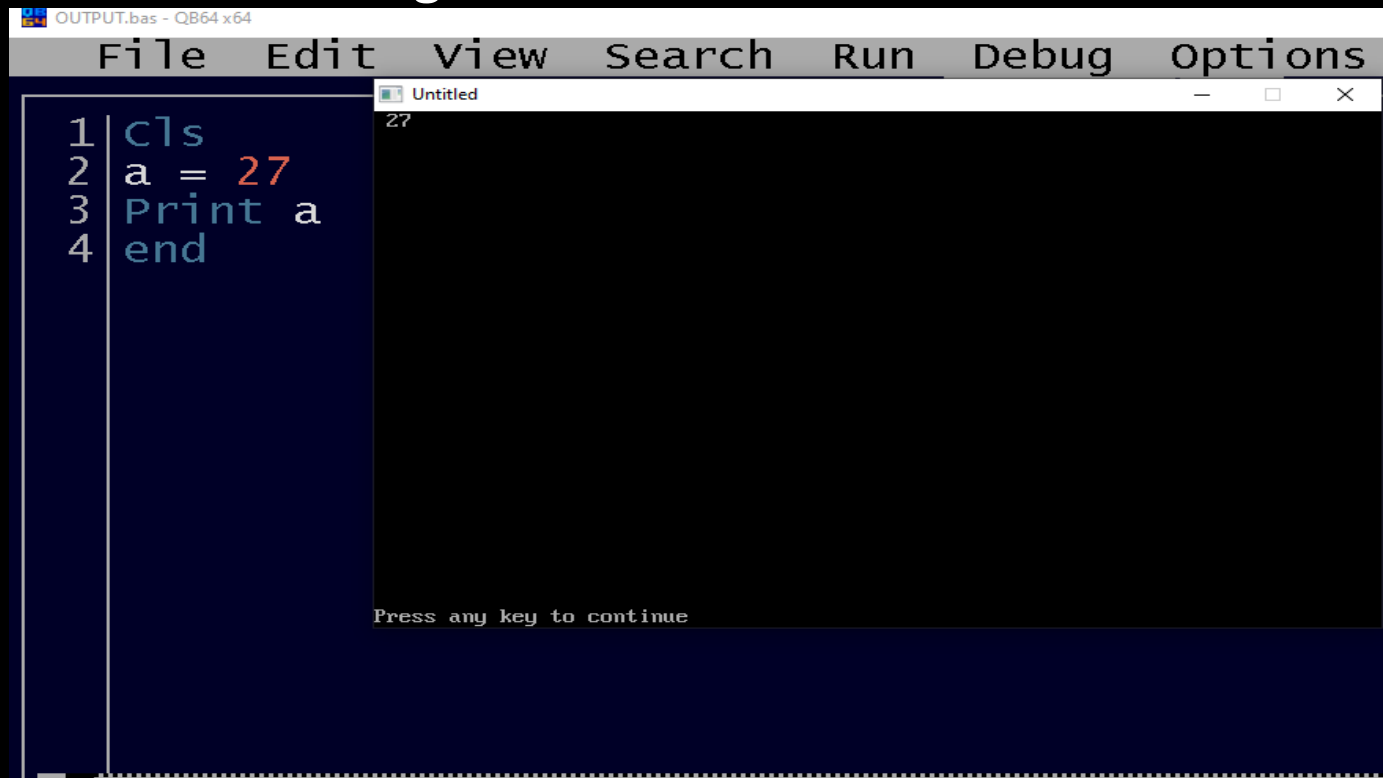
Untitled

? 27
You have Entered number is -: 27

4. End Command

- Used to program end.
- After use end command no any statement execute.
- Syntax :- END

Q1. Print 27 using variables.



The screenshot shows a QB64 IDE window titled "OUTPUT.bas - QB64 x64". The menu bar includes File, Edit, View, Search, Run, Debug, and Options. The code editor on the left contains the following BASIC code:

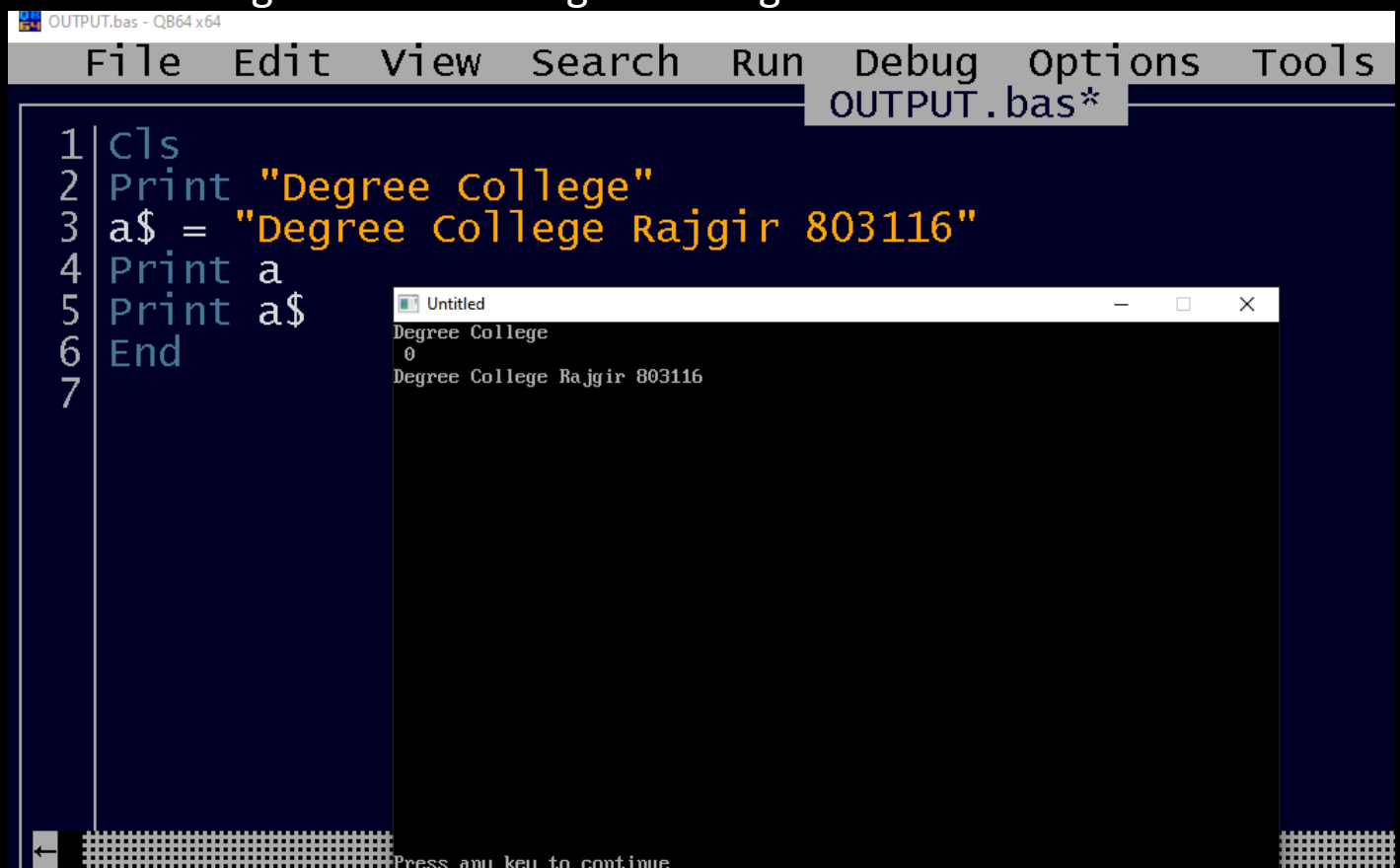
```
1 Cls
2 a = 27
3 Print a
4 end
```

The output window on the right, titled "Untitled", displays the number "27". At the bottom of the output window, it says "Press any key to continue".

Q2. Print String variable & String Constant

e.g. – String Variable - Degree College Rajgir 803116

String Constant - "Degree College"



The screenshot shows a QB64 IDE window titled "OUTPUT.bas - QB64 x64". The menu bar includes File, Edit, View, Search, Run, Debug, Options, and Tools. The code editor on the left contains the following BASIC code:

```
1 Cls
2 Print "Degree College"
3 a$ = "Degree College Rajgir 803116"
4 Print a
5 Print a$
6 End
7
```

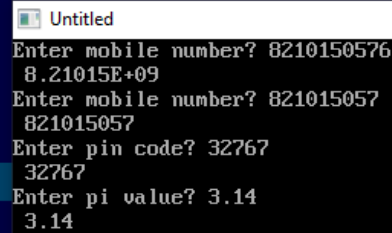
The output window on the right, titled "Untitled", displays the following output:

```
Degree College
0
Degree College Rajgir 803116
```

At the bottom of the output window, it says "Press any key to continue".

Q. Write a program to use DIM command /statement and print integer value, long integer value and float value.

```
Cls
Rem integer range (2byte) 32,767
Dim a, c As Integer
Dim b As Long
Dim d As Single
Input "Enter mobile number"; a
Print a
Input "Enter mobile number"; b
Print b
Input "Enter pin code"; c
Print c
Input "Enter pi value"; d
Print d
```



```
Untitled
Enter mobile number? 8210150576
8.21015E+09
Enter mobile number? 821015057
821015057
Enter pin code? 32767
32767
Enter pi value? 3.14
3.14
```

Program Using Mathematical Operators

Operator	Meaning/use	Example
+	Addition	Print 5 + 2 = 7
-	Subtraction	Print 5 - 2 = 3
*	Multiplication	Print 5 * 2 = 10
/	Division (decimal)	Print 5 / 3 = 2.5
\	Integer division	Print 5 \ 3 = 2
Mod	Remainder	Print 5 mod 2 = 1
^	Power	Print 5 ^ 2 = 25

❖ **3. Write a program add any two number in qbasic?**

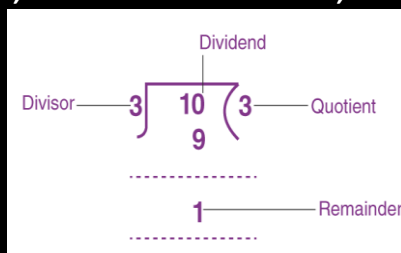
```
OUTPUT.bas - QB64 x64
File Edit View Search Run Debug Options Tools
OUTPUT.bas*

1 Cls
2 Print "Goverment Degree College,Rajgir"
3 Print "======"
4 Print "Add Two Number"
5 Input "Enter first number "; a
6 Input "Enter Second number "; b
7 Print "Sum ="; a + b
8 End
```



Divides Two numbers and prints the remainder.

e.g.- $10 \text{ MOD } 3 = 1$, $5 \text{ Mod } 2 = 1$, $10 \text{ Mod } 5 = 0$



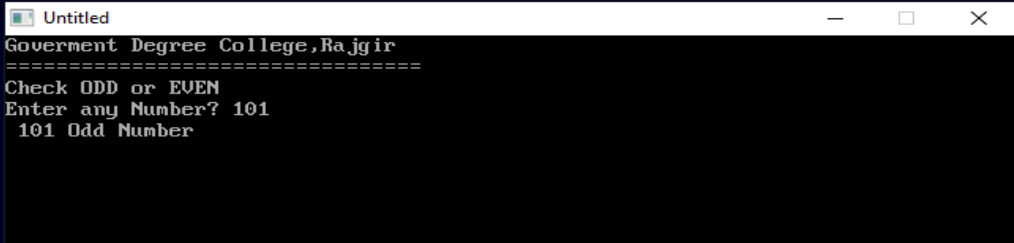
Q1. Write a program Take input any number and check this number is Odd OR Even.

➔ I know that 1 is Odd number and 0 is even number.

Also i know that any number divided by 2 then remainder will be 1 or 0.

```
OUTPUT.bas - QB64 x64
File Edit View Search Run Debug Options Tools
OUTPUT.bas*

1 Cls
2 Print "Goverment Degree College,Rajgir"
3 Print "======"
4 Print "Check ODD or EVEN"
5 Input "Enter any Number"; a
6 If a Mod 2 = 1 Then Print a; "Odd Number" Else Print a; "EVEN Number"
7 End
```



❖ Program Using Relational Operators

(Used to compare two values. Result is given as TRUE /FALSE.

E.g =,<>,>,<,>=,<=,BEDMASRule,WhereE-Exponentiation i.e.Power)

Operations	Operators	Example	Result
Equal to	=	6=6	TRUE
		6=10	FALSE
Not equal to	<>	5<>4	TRUE
		5<>5	FALSE
Greater than	>	12>5	TRUE
		15>27	FALSE
Less than	<	5<6	TRUE
		8<6	FALSE
Greater than equal to	>=	5>=5	TRUE
		5>=4	TRUE
Less than equal to	<=	7<=7	TRUE
		7<=5	FALSE

Q1. Write a program to check if a number is greater than 120 ?

Q2. Write a program to check if a number is smaller than 120 ?

```

1 Cls
2 Print "Government Degree College,Rajgir"
3 Print "=====
4 Print "Print Greater & Smallest number compair with 120"
5 Input "Enter any number"; a
6 If a > 120 Then Print a; "number is Greater than 120"
7 If a < 120 Then Print a; "number is smaller than 120"
8 End

```

```

Untitled
Government Degree College,Rajgir
=====
Print Greater & Smallest number compair with 120
Enter any number? 100
100 number is smaller than 120

```

OR

```

1 Cls
2 Input "Enter any number = "; a
3 If a > 120 Then GoTo yes
4 GoTo no
5 yes:
6 Print "yes "; a; " is a grater than 120"
7 GoTo ending:
8 no:
9 Print "no" ; a ; " is a smaller than 120"
10 ending:
11 End

```

```

Untitled
Enter any number = ? 80
no 80 is a smaller than 120

```

❖ Program Using logical Operators

Combine two statements and give us a result in TRUE/False

Operator	Operation	Example X=10	Result
AND	Returns FALSE if one of the statements is false.	X>5 AND X<8	FALSE
		T AND F	
OR	Returns TRUE if one of the statements is TRUE.	X>5 OR X<8	TRUE
		T OR F	
NOT	Returns TRUE if the statement is FALSE and FALSE if statement is TRUE	NOT X>5	FALSE
		NOT T	



NOT A = Q

A	Q
0	1
1	0



A OR B = Q

A	B	Q
0	0	0
0	1	1
1	0	1
1	1	1



A AND B = Q

A	B	Q
0	0	0
0	1	0
1	0	0
1	1	1

Q1. Write a Program to check the number between 100 to 200

```

OUTPUT.bas - QB64 x64
File Edit View Search Run Debug Options Tools
OUTPUT.bas*
1 Cls
2 Print "Goverment Degree College,Rajgir"
3 Print "======"
4 Print "Logical operators"
5 Input "Enter any number"; a
6 If a > 100 And a < 200 Then Print "True Number"; a; "is available to b/w 100 to 200"
7 If a < 100 Then Print "False Number"; a; "is not available to b/w 100 to 200"
8 If a > 200 Then Print "False Number"; a; "is not available to b/w 100 to 200"
9 End

Untitled
Goverment Degree College,Rajgir
=====
Logical operators
Enter any number? 111
True Number 111 is available to b/w 100 to 200

```

=====HIRA KUMAR=====